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**6th Sem / Mechanical Engineering, GE, Mech. Engg.
(Fabrication Tech.)
Sub : Automobile Engineering**

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 A diesel engine as compared to petrol engine (both running at full load) is (CO1)

- a) Less efficient b) More efficient
- c) Equally efficient d) None of these

Q.2 The problems caused by the wheel imbalance are (CO2)

- a) Hard steering and hard ride
- b) Poor acceleration and hard steering
- c) Steering wheel vibrations and uneven tyre wear
- d) Poor acceleration and reduced fuel efficiency

Q.3 The torque converter uses _____ to transfer torque. (CO3)

- a) Air
- b) Auto matic transmission fluid (ATF)
- c) Gear
- d) Steel belt

- Q.4 The instrument used to check specific gravity of acid in a battery is (CO4)
- a) Hydrometer b) Hygrometer
c) Anemometer d) Multimeter
- Q.5 The petrol engine works on (CO4)
- a) Otto cycle b) Diesel cycle
c) Carnot cycle d) Diesel cycle
- Q.6 The starter motor is driven by (CO5)
- a) Chain drive b) Gear drive
c) Flat belt drive d) V-belt drive
- Q.7 The component in the radiator of an automobile that increases the boiling point of water is (CO5)
- a) Drain plug b) Water jacket
c) Vacuum valve d) Pressure cap.
- Q.8 If the engine coolant leaks into the engine oil, then engine oil (CO6)
- a) Appears milky b) Become foamy
c) Turns black d) None of these
- Q.9 A clutch is usually designed to transmit maximum torque which is (CO6)
- a) Equal to the maximum engine torque
b) 80% of the maximum engine torque
c) 150% of the maximum engine torque
d) None of these
- Q.10 In a four stroke engine for each crankshaft revolution, the camshaft revolves. (CO6)
- a) One half turn b) One turn
c) Two turns d) Four turns

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 The petrol engines are also known as _____. (CO1)
- Q.12 The problems caused by the wheel imbalance are _____. (CO2)
- Q.13 The torque converter uses _____ to transfer torque. (CO3)
- Q.14 The instrument used to check specific gravity of acid in a battery is _____. (CO3)
- Q.15 The petrol engine works on _____. (CO4)
- Q.16 The starter motor is driven by _____. (CO5)
- Q.17 The component in the radiator of an automobile that increases the boiling point of water is _____. (CO5)
- Q.18 If the engine coolant leaks into the engine oil, then engine oil _____. (CO6)
- Q.19 A clutch is usually designed to transmit maximum torque which is _____. (CO6)
- Q.20 In a four stroke engine for each crankshaft revolution, the camshaft revolves _____. (Co6)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Write a short note on 6x4 wheel drive. (CO1)
- Q.22 Write a short note on electric vehicles. (CO1)
- Q.23 Classify gear box. (CO2)
- Q.24 Write the functions of transmission system. (CO2)
- Q.25 What is the necessity of gear box? (CO2)

- Q.26 Write the advantages of single plate clutch. (CO2)
- Q.27 Write the function of tyres. (CO2)
- Q.28 Write the principle of hydraulic power steering. (CO3)
- Q.29 Explain, in brief power steering. (CO3)
- Q.30 Classify brakes according to mode of operation. (CO4)
- Q.31 Write the advantages of hydraulic brake system. (CO4)
- Q.32 Explain air suspension system with the help of neat sketch. (CO5)
- Q.33 Write the functions of suspension system. (CO5)
- Q.34 Explain trickle charging of battery. (CO6)
- Q.35 Write the functions of electrical system in an automobile. (CO6)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Define automobile. Name its different components. Explain briefly. (CO1)
- Q.37 Explain the construction and working of single plate clutch with the help of neat sketch. (CO2)
- Q.38 Explain Davis steering mechanism with the help of neat sketch. (CO3)

Note : Course Outcomes (CO) mentioned in the question paper is for official purpose only.